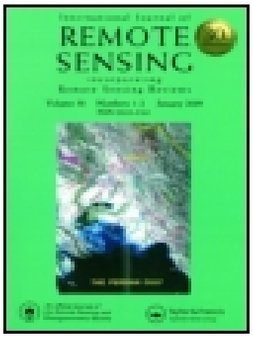

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A comparison of machine and deep-learning algorithms applied to multisource data for a subtropical forest area classification

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ABSTRACT

This work explores the integration of airborne Light Detection and Ranging (LiDAR) data and WorldView-2 (WV2) images to classify the land cover of a subtropical forest area in Southern Brazil. Different deep and machine learning methods were used: one based on convolutional neural network (CNN) and three ensemble methods. We adopted both pixel- (in the case of CNN) and object-based approaches. The results demonstrated that the integration of LiDAR and WV2 data led to a significant increase (7% to 16%) in accuracies for all classifiers, with kappa coefficient (κ) ranging from 0.74 for the random forest (RF) classifier associated with the WV2 dataset, to 0.92 for the forest by penalizing attributes (FPA) with the full (LiDAR + WV2) dataset. Using the WV2 dataset solely, the best κ was 0.81 with CNN classifier, while for the LiDAR dataset, the best κ was 0.8 with the rotation forest (RotF) algorithm. The use of LiDAR data was especially useful for the discrimination of vegetation classes because of the different height properties among them. In its turn, the WV2 data provided better performance for classes with less structure variation, such as field and bare soil. All the classification algorithms had a nearly similar performance: the results vary slightly according to the dataset used and none of the methods achieved the best accuracy for all classes. It was noticed that both datasets (WV2 and LiDAR) even when applied alone achieved good results with deep and machine learning methods. However, the advantages of integrating active and passive sensors were evident. All these methods provided promising results for land cover classification experiments of the study area in this work.

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1. Introduction

Tropical forests are among the most complex ecosystems on Earth, playing crucial roles in biodiversity conservation and in ecology dynamics at a global scale (Zhang Z. et al. 2016). The Atlantic Rain Forest of Brazil is a highly diverse, ecologically important, but threatened tropical forest. According to Joly, Metzger, and Tabarelli (2014), the high levels of

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deforestation and fragmentation to which this forest has been exposed have affected its diverse biota. As a result, its forest remnants are sparsely distributed and present different succession stages, covering less than 16% of its original area (Ribeiro et al. 2009).

The Mixed Ombrophilous Forest (MOF) or Araucaria Moist Forest is one of the main phytophysiognomies of the Atlantic Rain Forest in Southern Brazil (Higuchi et al. 2012). It is characterized by the presence of *Araucaria angustifolia* (Parana´s pine tree), henceforth araucaria, as an emerging species over the canopy, acting as an indicator of this forest phytophysiognomy. MOF is regarded as one of the Brazil´s most threatened forest types, because only 24% of its original cover is still standing (Vibrans et al. 2013). Its remnants are composed mainly of small isolated patches (<50 ha) of secondary forests in initial (SS_1) and intermediate succession stages (SS_2) (Ribeiro et al. 2009). Considering this, accurate mapping of both its few remaining forest remnants and endangered tree species is crucial to support future management practices, monitoring and conservation strategies.

In this respect, field surveys are efficient; however, they tend to be time-consuming and costly to execute at a large scale, systematically and in areas of difficult access. These restraints have encouraged initiatives based on remote sensing as an effective means to map land use/land cover, vegetation succession stages (Lu et al. 2014; Sothe et al. 2017) and tree species, including the endangered ones (Pouteau et al. 2012; Omer et al. 2015).

For forest types or tree species classification, data with both high spatial and spectral resolution are desirable (Clark and Roberts 2012; Féret and Asner 2013). There is currently a large variety of sensors with a wide range of spatial and spectral resolutions. Because airborne cameras are limited in their spectral resolution and airborne hyperspectral sensors are disadvantageous due to their high costs and still limited availability, some of the multispectral satellite data have proven to be the most viable alternative for forest resource mapping. Satellites such as WorldView-2 (WV2) provide very high spatial resolution and four new spectral bands (coastal, yellow, red edge, and near infrared 2) besides the traditional visible and near-infrared (VNIR) bands (Immitzer, Atzberger, and Koukal 2012). Nevertheless, a limitation of these data is that passive optical remote sensing data neglect the three-dimensional characteristics of ground objects, which is important in terms of land cover classification accuracy (Fieber et al. 2013). Especially in the case of different forest types, the similarity of spectral signatures increases the challenges for successful mapping from optical remote sensing data alone (Leckie et al. 2005).

In some cases, the described problems can be reduced by integrating canopy structural information from other remote sensing sources, notably from airborne Light Detection and Ranging (LiDAR) (Castillo et al. 2012; Ghosh et al. 2014). Despite providing accurate vertical information, LiDAR has limited spatial coverage, mostly due to its cost, especially for data acquisition over large areas (Stojanova et al. 2010). In this way, the incorporation of additional data sources, such as passive optical remote sensing data, may mitigate the cost of data acquisition, and improve the accuracy and efficiency of LiDAR applications (Debes et al. 2014). Especially in the case of different forest classes, the combined use of passive multispectral data with active sensors data has shown to be a promising approach for land cover classification, since they provide complementary information (spectral response and vertical structure, respectively) on the targets of interest (Dalponte, Bruzzone, and Gianelle 2012), thus optimizing the strengths of both data sources (Reese et al. 2014). Several studies have proved the benefit of integrating optical images and airborne LiDAR for forest applications (Ke, Quackenbush, and Im 2010;

Dalponte, Bruzzone, and Gianelle 2012; Ghosh et al. 2014; Reese et al. 2014; Baldeck et al. 2015; Zhang Z. et al. 2016). However, only a few of them have investigated the use of LiDAR data alone (e.g., Heinzel and Koch 2011; Castillo et al. 2012).

Besides the data, the choice of the classifier is also decisive for reliable land cover and land use mapping (Lu and Weng 2007). In this regard, many studies have demonstrated that machine learning algorithms are more accurate than the traditional classification techniques such as parametric algorithms, especially when the feature space is complex and the data to be classified present different statistical distributions (Mountrakis, Im, and Ogole 2011). The increasing use of these methods in remote sensing over recent years is due to different factors: the ability of these techniques to learn complex patterns, non-linear in many cases; the high generalization capacity of these algorithms, which makes it possible to apply them to incomplete or noisy databases; the possibility of incorporating a priori information in the analysis; and the capacity to handle data regardless of their statistical distribution (Mas and Flores 2008; Mountrakis, Im, and Ogole 2011; Maxwell, Warner, and Fang 2018). This latter advantage renders available the addition of data from multiple sensors, auxiliary variables, and even categorical variables (Rodriguez-Galiano and Chica-Rivas 2014).

Among the most widely used machine learning methods are artificial neural networks (ANN) (Mas and Flores 2008), and ensemble methods such as random forest (RF) (Breiman 2001). Other ensemble methods have emerged from RF, for example rotation forest (RotF) (Rodríguez, Kuncheva, and Alonso.2006) and forest by penalizing attributes (FPA) (Adnan and Islam 2017). These methods have not yet been fully explored for the classification of forest areas. Deep-learning is another concept of ANN that has emerged as the next-generation learning technique with great potential for classifying remote sensing images (Yue et al. 2015; Li, Zhang, and Shen 2017; Li et al. 2017; Wang et al. 2017; Gao, Samsung, and Jia 2018). Among the deep-learning algorithms, the convolutional neural network (CNN), first proposed by Fukushima (1980), has become outstanding, among other reasons, because it allows handling data in two or three dimensions, which is the case of remote sensing images classification (Li et al. 2017).

Although there have been many studies investigating the effectiveness and efficiency of different machine learning classifiers, such as support vector machine (SVM), ANN and RF (Song, Duan, and Jiang 2012; Ghosh et al. 2014; Raczko and Zagajewski 2017), most of them have been accomplished using pixel-based approaches. However, the presence of noise, differences in lighting conditions and spectral variability within the canopies and among species when we deal with high spatial resolution images may negatively affect the classification results at a pixel scale (Yu et al. 2006). In turn, at an object scale, segments are used as classification units, reducing the negative effects of spectral variability of pixels (Heinzel and Koch 2011). Considering that previous investigations have shown that Geographic Object-Based Image Analysis (GEOBIA) approaches improve classification accuracies in land cover image classification (Ma et al. 2017), there has been an increasing interest in comparing different machine learning classifiers using GEOBIA methods (Qian et al. 2014). In this respect, Qian et al. (2014) compared four machine learning classifiers (SVM, normal Bayes, *k*-nearest neighbour and decision tree) adopting an object-based approach to classify an urban landscape. However, there is still a lack of studies comparing recent machine learning as well as deep-learning methods using GEOBIA.

In light of the above, the aim of this work is to analyse the integration of WV2 multispectral data with airborne LiDAR data for the semiautomatic land cover mapping of a subtropical forest area of MOF in the Atlantic Rain Forest of Brazil. The specific goals are: i) to analyse the performance attained in classifications relying on the use of WV2 and LiDAR data alone and compare them to classifications results integrating both datasets and; ii) to test and compare different machine learning methods, both those more often used, such as RF, as well as barely explored algorithms in forest applications, such as RotF and FPA, and a deep-learning method, CNN. These algorithms were specifically chosen because they are being increasingly used for land cover mapping (e.g., Duro, Franklin, and Dubé 2012; Rodriguez-Galiano et al. 2012; Xia et al. 2013; Lu et al. 2014; Kavzoglu, Colkesen, and Yomralioglu 2015; Driss, Soua, and Akil 2017; Raczko and Zagajewski 2017; Sothe et al. 2017; Yu, Jia, and Xu 2017) and yet have not been compared with one another exhaustively.

The main contributions of this work are:

(a) A performance assessment of combining multispectral and LiDAR data to classify forest areas (i.e. successional forest stages, endangered and exotic tree species), the target application.
(b) A comparison of distinct multiclassifier approaches based on decision trees for object-based classification of forest areas.
(c) An evaluation of a fully-connected CNN for pixel-wise classification of forest areas.
(d) A comparison between object-based (OBIA) approaches with pixel-wise classification provided by deep CNNs.

2. Material and methods

The methodological framework developed for this research is shown in Figure 1. The common steps for all classifiers comprised the pre-processing of both data (WV2 and LiDAR), feature extraction at the pixel level, compilation of datasets and collection of

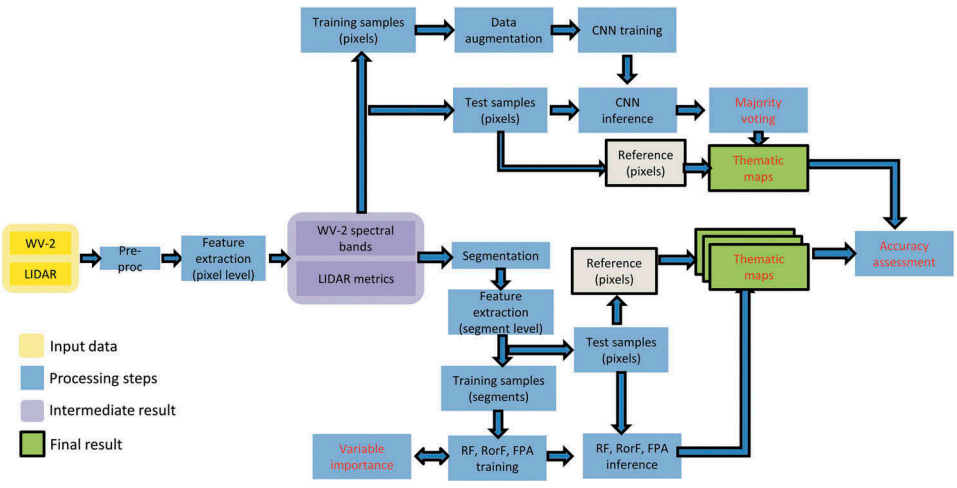


Figure 1. Methodological framework developed in this study.

training samples. For the ensemble methods, a segmentation was performed prior to the classification. Then, the spectral means corresponding to each layer of the training segments were used for the classification process. In this step, the importance of the variables was also evaluated. For the CNN, a data augmentation was initially applied to the training samples and in the sequence a pixel-based classification was executed. The final CNN result was aggregated into segments using a majority-vote procedure and the segmentation result also adopted by the other classifiers. All the classification results were evaluated, and then final thematic maps were produced.

2.1. Study area

The study area covers 64 ha and is located in the municipality of Painel, Santa Catarina state, south of Brazil (Figure 2). The area belongs to the Atlantic Rain Forest biome and shelters the MOF phytophysiognomy. It is located near the Lava-Tudo River at an elevation above sea level of 1,000 m. The climate, according to Köppen's classification, is temperate oceanic climate (Cfb), moist mesothermal with no clearly defined dry season, mild summers, often and severe freezes in winter, and mean temperature of less than 12° C. The area has an average annual rainfall of 1,543 mm.

2.2. Input data

The WV2 scene, with eight multispectral bands and one panchromatic band (Table 1), used in this work was acquired in May of 2012. We acquired a cloud-free scene; however, shadows of clouds are present in some parts of the images (Figure 2(b)). Initially, these data were converted to radiance images, using the Radiometric Calibration tool of Environmental for Visualizing Images (ENVI) 5.0. Next, the images were once again

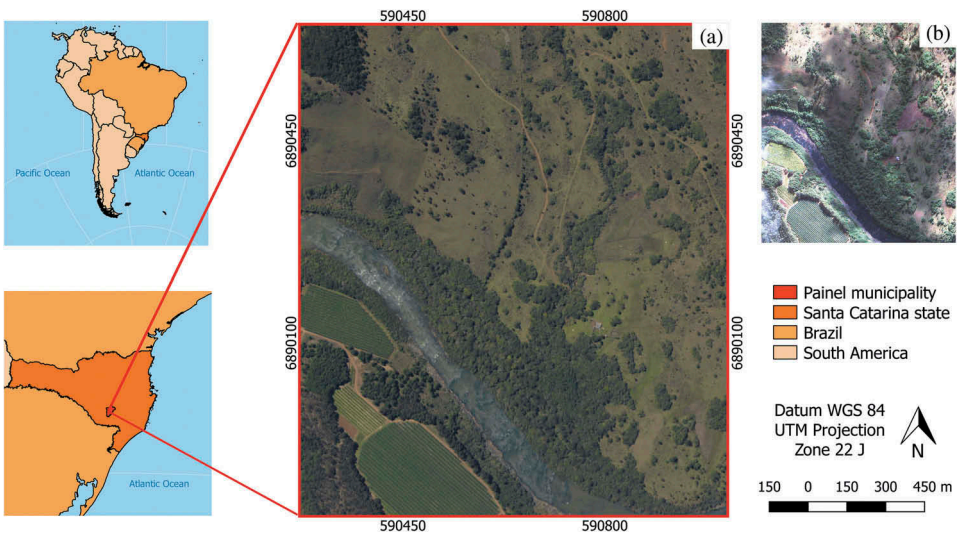


Figure 2. Study area location. a) Airborne System for Acquisition and Post-processing Images (SAAPI) orthoimage (the red square indicates the study area boundaries); b) the corresponding WV2 scene – Red 5 Green 3 Blue 2 composition.

Table 1. Description of features used in this study.

Group	Feature name	Description	Equation
Spectral bands	Panchromatic	Wavelength 0.46–0.80 µm	
	Coastal	Wavelength 0.40–0.45 µm	
	Blue	Wavelength 0.45–0.51 µm	
	Green	Wavelength 0.51–0.58 µm	
	Yellow	Wavelength 0.59–0.63 µm	
	Red	Wavelength 0.63–0.69 µm	
	Red-edge	Wavelength 0.71–0.75 µm	
	Near Infrared 1	Wavelength 0.77–0.90 µm	
	Near Infrared 2	Wavelength 0.86–1.04 µm	
VIs	NDVI	Normalized Difference Vegetation Index	$\mathrm{NDVI} = \frac{\rho_{\mathrm{NIR}}-\rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}}+\rho_{\mathrm{Red}}}$
	SR	Simple Ratio	$\mathrm{SR} = \frac{\rho_{\mathrm{NIR}}}{\rho_{\mathrm{Red}}}$
	OSAVI	Optimized Soil Adjusted Vegetation Index	$\mathrm{OSAVI} = \frac{\rho_{\mathrm{NIR}}-\rho_{\mathrm{Red}}}{\rho_{\mathrm{NIR}}+\rho_{\mathrm{Red}+0.16}}$
	$\mathrm{NDVI}_{\mathrm{Red\text{-}edge}}$	Red-edge Normalized Difference Vegetation Index	$\mathrm{NDVI}_{\mathrm{Red\text{-}edge}} = \frac{\rho_{\mathrm{Red-edge}}-\rho_{\mathrm{Red}}}{\rho_{\mathrm{Red-edge}}+\rho_{\mathrm{Red}}}$
GLCM	SM	Angular second moment	$\mathrm{SM} = \sum_{i,j=0}^{N-1}\left\{p(i,j)^2\right\}$
	Con	Contrast	$\mathrm{Con} = \sum_{i,j-0}^{N-1} p(i,j)(i-j)^2$
	Var	Variance	$\mathrm{Var} = \sum_{i,j=0}^{N-1}(i-\mu)^2 p(i,j)$
	Hom	Homogeneity	$\mathrm{Hom} = \sum_{i,j=0}^{N}\frac{p(i,j)}{1+(1-j)^2}$
	Cor	Correlation	$\mathrm{Cor} = \sum_{i,j=0}^{N} p(i,j)\frac{[(i-\mu i)(j-\mu j)]}{\sqrt{(\sigma^2 i)(\sigma^2 j)}}$
	Ent	Entropy	$\mathrm{Ent} = \sum_{i,j-0}^{N-1} p(i,j)(-\ln p(i,j))$
	Dis	Dissimilarity	$\mathrm{Dis} = \sum_{i,j-0}^{N-1} p(i,j)\lvert 1-j\rvert$
	Mean	Textural mean	$\mathrm{Mean} = \sum_{i,j=1}^{N} i(p(i,j))$
CHM	CHM	Canopy height model	CHM = DSM − DTM
LiDAR metrics	ipcumzq90	percentage of intensity returned below the 90th percentile of height	
	imean	intensity mean of all points	
	imax	maximum of all intensity points	
	zq10	10th percentile of height distribution	
	zkurt	kurtosis of height distribution	
	zskew	skewness of height distribution	
	zmean	mean of all height points	

Note: *i* is the row number; *j* is the column number; *p(i,j)* is the normalized value in the cell *i,j; N* is the number of rows or columns; *µ* is the mean value of *p*; *σ* is the standard deviation of *p*.

converted to surface reflectance using the Fast Line-of-sight Atmospheric Analysis of Spectral Hypercubes (FLAASH). After atmospheric correction, the WV2 multispectral bands with 2 m of spatial resolution were pansharpened with the WV2 panchromatic band, with 0.5 m, by means of the Gram-Schmidt method. We opted for this method because it does not improve or decrease the classification accuracy but it is important if high spatial detail is needed (Bruzzone et al. 2006). In this study, the spatial resolution is important, since we considered a class containing only a single tree species. Thus, a spatial resolution of 2 m would reduce the possibility to detect this class correctly.

Finally, the WV2 scene was orthorectified based on the rational polynomial coefficients, using parameters provided by the metadata and a Digital Surface Model (DSM) from the LiDAR data (described below). For geocoding the data, an orthoimage with a spatial resolution of 0.39 m acquired by the Airborne System for Acquisition and Post-processing Images (SAAPI) was used as reference together with 15 ground control points. Further details about the WV2 sensor can be found in Digital Globe (2013).

The LiDAR data, in their turn, were acquired by the Leica ALS-60 sensor onboard of a flight undertaken in January of 2011, with an average density of 7 points/m^2 and intensity data at the

1,064 nm wavelength. Firstly, outliers were removed and then the DSM and the Digital Terrain Model (DTM) were generated using the Adaptive Triangular Irregular Network (TIN) filter (Axelsson 2000) in the LAStools software (Isenburg 2012). Finally, the Canopy Height Model (CHM) was extracted by the subtraction between the DSM and DTM.

2.3. Feature extraction and variable importance

The following stage consisted of extracting features from both datasets (Table 1). The use of ancillary data with multispectral images has shown to be advantageous in classification processes with machine learning algorithms (Ke, Quackenbush, and Im 2010; Pu and Landry 2012; Kavzoglu, Colkesen, and Yomralioglu 2015). So, for WV2, eight texture metrics were calculated based on four multispectral bands (blue, green, red and near-infrared 1) using the Grey-Level Co-occurrence Matrix (GLCM) proposed by Haralick, Shanmugam, and Dinstein (1973). To compute these metrics, the window size was set to 5×5 considering the southwest direction and level of quantization of 64 bits.

The four VIs mentioned in Table 1 were calculated and integrated to the WV2 dataset. For the CNN classifier, only the original bands of WV2 were employed, since this classifier can automatically extract useful features from the training set (Zhang L. et al. 2016).

In the case of the LiDAR data, besides the CHM, seven height and intensity-derived metrics described in Table 1 were extracted using the R language (R Development Core Team 2013) and the lidR package (Roussel et al. 2018). Some of these features were used for tree species or succession forest stages classifications (e.g., Ørka, Næsset, and Bollandsås 2009; Castillo et al. 2012; Dalponte, Bruzzone, and Gianelle 2012; Reese et al. 2014), because they describe the structural characteristics of the forest (Dalponte, Bruzzone, and Gianelle 2012). The spatial resolution of all LiDAR-derived metrics was set to 1m in order to avoid the generation of areas with missing values. Subsequently, they were resampled to 0.5 m in order to make their spatial resolution similar to the WV2 pansharpened data . Three datasets were thus created and they are described in Table 2.

In order to check the importance of each feature in the classification, we used two feature ranking methods: the Jeffries-Matusita (JM) distance and the mean decrease in the Gini index (MDG) of the RF algorithm. The JM has upper and lower bounds that vary between 0 and $\sqrt{2}$ ($\approx$1.414), with the higher values indicating the total separability of the class pairs in the bands being used (Richards 1993).

The MDG is a feature importance statistic produced by the RF averaging Gini indices of the individual trees (Breiman 2001). A higher MDG indicates higher variable importance; conversely, the least important variable will have the smallest MDG values (Louppe et al. 2013).

Table 2. Description of the classification experiments according to the features used in the classification process.

Dataset/Group	Spectral bands	VIs	GLCM	CHM	LiDAR metrics	Total features
WV2	8	4*	32*			44
LiDAR				1	7	8
full	8	4	32	1	7	52

*Except for CNN classifier. A better description of the features is given in Table 1.

2.4. Image segmentation

At this point, two distinct processes were employed: one for the CNN classifier, comprising a pixel-based classification grouped into segments using a majority-vote procedure; and the other one associated with an object-based classification employing the RF, RotF and FPA classifiers (as shown in Figure 1). The multiresolution segmentation (Baatz and Schäpe 2000), available in the eCognition software, was applied to a dataset composed of the original bands of WV2 and CHM. The multiresolution algorithm has been widely used for applications involving forest environments (e.g., Tsendbazar 2011; Whiteside 2011; Singh et al. 2015). This algorithm starts with seed pixels and progressively grows regions by iteratively including adjacent pixels until a threshold of expansion or stopping criteria is met.

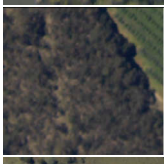
In this study, the segmentation parameters were optimized using the Segmentation Parameter Tuner (SPT) software (Feitosa et al. 2010; Achanccaray et al. 2014), since it concerns all the multiresolution segmentation parameters, not only the optimal scale parameter. First, reference segments were manually delineated over the image. The number of references per class varied between 5 (water) and 20 (araucaria). We set more references for araucaria to ensure that this class would not be undersegmented, since it comprises scattered trees over the field. Based on these references, the optimization function (in this case, Precision and Recall) interacts with the segmentation algorithms to determine the optimal set of parameters. The second step was to select the optimization method. We opted for a hybrid strategy (Quirita et al. 2016), in which the Differential Evolution is firstly applied to identify a promising initial solution, and then refined by the Nelder-Mead function (Nelder and Mead 1965). The Differential Evolution parameters were set to 10 for population and 4 for generation, associated with the DE/best/1/exp strategy (details in Storn and Price 1997). For Nelder-Mead, the number of iterations was set to 200. The optimization was run three times in order to avoid a biased selection of a minimum local as seed point and to ensure that the algorithm converges prior to that (Quirita et al. 2016). The best result of these runs reached a Precision and Recall value of 0.21, indicating a good balance between over and undersegmentation. The closer the value is to zero, the better the spatial match between references and the resulting segments and thus, the larger the overlapping area between them (Pont-Tuset and Marques 2013).

The parameters automatically chosen were visually refined during the segmentation process executed in eCognition. In the final parameters, the scale factor was set to 15, the shape factor to 0.2 and the smoothness factor to 0.5. We opted for an oversegmentation result, which is more appropriate than undersegmentation, which corresponds to the presence of distinct targets within the same segment (Baatz and Mimler 2007).

2.5. Samples collection

After the segmentation, training samples were collected for eight classes (Table 3) based on the SAAPI orthoimage and field observations. The samples collected cover the spectral variability of each class and consider the lighting variation in the WV2 scene. For the ensemble classifiers, the samples and all remaining segments were exported as a shapefile containing all the features previously indicated in Table 1. For CNN, the samples were converted in a tagged image file format (TIFF) file and were regarded as pixels instead of segments.

Table 3. Land cover classes used in the classification process and their respective number of segments and pixels samples. Figure showing the classes in an Airborne System for Acquisition and Post-processing Images (SAAPI) orthoimage in true colour composition.

Class description	Legend	Samples (segments)	Samples (pixels)	Figure
Natural vegetation in a secondary intermediate stage of recovery	SS_2	61	11,890	
Natural vegetation in a secondary early stage of recovery	SS_1	59	6,090	
Reforestation (exotic species *Pinus* sp.)	reforestation	63	5,558	
Araucaria angustifolia (dominant and endangered tree species in the region)	araucaria	40	3,185	
Agriculture (crops)	crop	68	8,193	
Bare soil	bare soil	21	1,291	
Fields	fields	109	13,142	
Water bodies	water	31	3,345	

2.6. Classification techniques

2.6.1. Ensemble classifiers

RF was developed as a way to improve the accuracy of classification and regression trees by combining a large number of random trees. In this case, each tree contributes with only one class vote for each instance, and the final classification is determined by the

majority of the votes of all forest trees (Hastie, Tibshirani, and Friedman 2009). The trees in RF are created by drawing a subset of training data through a bagging approach. The bagging randomly selects about two thirds of the samples from training data to train these trees. This means that the same sample can be used in a training subset several times, while others may not be selected in a particular subset (Belgiu and Dragut 2016). In this study, this classifier was preliminarily tested using a 10-fold cross-validation procedure with 100, 500 and 1,000 trees. It was found that increasing the number of trees from 100 to 500 led to a small improvement in the classification result, while changing from 500 to 1,000 trees considerably increased the processing time with no corresponding increase in accuracy. In face of this, we opted to use 500 trees in all experiments. Regarding the *mtry* parameter, the default value was kept, which corresponds to the square root of the total number of features used in each experiment (Breiman et al. 2001).

RotF, proposed by Rodriguez et al. (2006), is related to the RF algorithm. The main idea of this method is to provide diversity among classifiers in an ensemble by designing each forest tree to operate on a random rotation of the original feature space. In this method, bootstrap samples are taken as training set for the individual classifiers, as in bagging. The main heuristic is to randomly separate the input features into K subsets upon which parts of a rotation matrix are constructed. This corresponds to a random rotation of the original input space. A decision tree is then trained on the rotated data. The procedure is repeated a number of times producing an ensemble of decision trees. For the inference procedure, the input data are first multiplied by the rotation matrices built during the training step and then subsequently classified by the corresponding tree (Kuncheva and Rodríguez 2007). Any decision tree can be used in the aforementioned schemes. In this work, we selected the decision tree C4.5 (Quinlan 1993) as the base classifier, 500 as the number of trees, 3 as the number of features in a subset, and 10 as the number of classifiers in the ensemble. This parameters setting was selected based on a cross-validation procedure, and since the performance varied very little, it was adopted for all datasets.

Proposed by Adnan et al. (2017), FPA aims to build a set of highly accurate decision trees by penalizing those features that participated in the latest tree in order to generate the subsequent trees. So, the probability that a feature is selected as split criterion in the design of the next tree of the assemble being built decreases as it has been selected in the design of prior trees. FPA also has a mechanism to gradually increase weights of the features that have not been tested yet in the generation of subsequent tree(s) (Adnan and Islam 2017). This method generates a bootstrap sample from the training dataset, and then generates a decision tree from the bootstrap sample using the weights of the features. The decision tree is built using the CART (classification and regression tree) algorithm (Breiman et al. 1985). The main parameter to be defined in the FPA is the number of trees, which, based on a 10-fold cross-validation, was set to 100 for all the datasets.

The ensemble classifiers were performed in software Waikato Environment for Knowledge Analysis (WEKA) (Hall et al. 2009) using the three databases (WV2, LiDAR, WV2+ LiDAR) created in Attribute-Relation File Format (ARFF).

2.6.2. Convolutional neural network

As the most representative supervised deep-learning model, the CNN (Fukushima 1980) is a trainable multilayer architecture composed of multiple feature-extraction stages.

Instead of using fully connected hidden layers as multilayer perceptron (MLP), the CNN introduces a special network structure, which consists of alternating layers named convolution and subsampling layers (Driss, Soua, and Akil 2017). The architecture of a CNN is designed to take advantage of the two-dimensional structure of the input image. In this research, a typical CNN architecture was applied, i. e. composed of feature-extraction stages, followed by a traditional fully connected layer and the softmax as the final classification layer.

To perform the CNN classification, instead of ARFF, the three datasets comprised the TIFF with: i) the original bands of the WV2 scene, ii) the LiDAR metrics, and iii) both of them together. The method was implemented in a Python environment using TensorFlow (Abadi et al. 2015). The evaluated network was designed to input a patch of 15 × 15 pixels and outputs a probabilistic vector of size equal to the number of classes, where the index location of the higher value indicates the most probable class. The network architecture consisted of five convolutional layers, three pooling layers, a fully-connected layer and a softmax classification layer (Figure 3). The number of kernels for each convolutional layer was 32, 32, 48, 48, 64, and 128 for the fully-connected layer and the learning rate of $10e^{-4}$. After every convolution operation and the fully-connected layer, a batch normalization was applied, followed by a leaky rectified linear unit (Leaky ReLU) activation function.

We set the Adam optimizer (Kingma and Ba 2015) parameters as default values. This method performs an efficient stochastic optimization that only requires first-order gradients with little memory requirements (Kingma and Ba 2015). To deal with overfitting, the network was trained using early stopping and dropout regularization of 0.35 after the fully-connected layer and before the softmax layer. The softmax, a function that computes class posterior probabilities (Gao, Samsung, and Jia 2018), was applied to the top layer of the proposed network to perform pixel wise classification upon the learned representative features. Details of softmax and Leaky RELU equations can be found in Goodfellow, Bengio, and Courville (2016) and Gao, Samsung, and Jia (2018).

The proportion of reference data for training and validation was about 50% and 50%. Considering the limitation of CNN regarding the number of training samples (Pasupa and Sunhem 2016), a data augmentation process was applied to increase the samples. To this end, the samples were replicated as the feature space was rotated and flipped in different directions until they reached 30,000 pixels per class. For inference, the CNN was applied to overlapping image patches to predict the class of their central pixel. Next, each query was spatially concatenated to obtain a thematic map at the same resolution of the input image.The final classifications of CNN were grouped into segments generated in the first

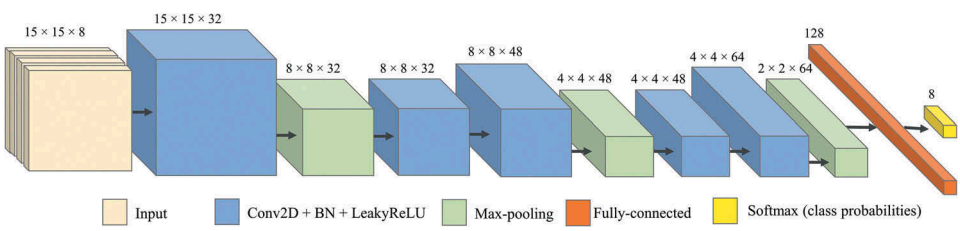


Figure 3. Convolutional neural network architecture used in this study.

stage using a majority-class rule procedure (Dalponte et al. 2013; Féret and Asner 2013; Ferreira et al. 2016). The final maps were evaluated following the same process as ensemble methods (c.f. section below).

2.7. Accuracy assessment

For the accuracy assessment, polygons meant for analyses were delimited in the image for each class by an independent expert photointerpreter, with the aid of the orthoimage and field observations. Based on these polygons, 50 pixels were randomly selected in each class. This procedure was repeated for each classification, so as to keep the independence among the test samples. The number of samples was defined according to Congalton and Green (2009), who recommend a minimum of 50 samples for maps of less than one million acres in size and fewer than 12 classes.

After the samples collection, error matrices were elaborated based on a cross-checking between the classified maps and their respective test samples. The kappa coefficient (κ) and conditional kappa per class (κ_c) were extracted from the matrices (Congalton and Green 2009). The z tests for the κ were executed with a level of significance of 5%, i.e., with a confidence interval of 95%. The value of the normal distribution z is obtained by the ratio of the difference between two distinct κ to the difference of their respective variances. When $z > 1.96$, the test is significant, the null hypothesis is rejected, and hence, there is significant statistical difference between the two values.

3. Results

3.1. Classification accuracies

Figure 4 shows the κ of each classification. The full dataset presented an increase of 7% to 16% in accuracies for all classifiers, with κ ranging from 0.88 with CNN and RF, to 0.92 with the FPA algorithm (Figure 5(a)). In the case of the WV2 dataset, the κ ranged from 0.74 for RF, to 0.81 for CNN (Figure 5(c)), while for the LiDAR dataset, the κ oscillated from 0.77 for FPA to 0.8 for RotF (Figure 5(b)). When comparing the WV2 with the LiDAR dataset separately, the performance of the classifiers did not vary much. We also can observe in Figure 4 that LiDAR dataset reached slightly better results with ensemble classifiers, while the CNN algorithm obtained its best results with the WV2 dataset.

Regarding the classifiers, the algorithm with the best performance varied according to the dataset. In the case of the full dataset, FPA attained the best result (κ of 0.92), although it did not significantly differ from the other classifiers, which reached similar results. When the performances of the algorithms were compared for the WV2 dataset, it was observed that CNN (κ of 0.81) were significantly superior to FPA and RF, not differing from RotF (κ of 0.77). For the LiDAR dataset, the RotF algorithm reached the best results (Figure 5(b)), however, it did not significantly differ from RF (κ of 0.79), CNN (κ of 0.78), and FPA (with κ of 0.77).

The κ_c showed that the accuracy of classifiers and datasets varied for each class, none of them reaching the best results for all classes (Table 4). Among the vegetation classes, the RotF classifier with the full dataset attained the best results for three classes: SS_2 and crop with κ_c of 0.98; and for araucaria (κ_c of 0.96). For SS_1, the best result was reached with

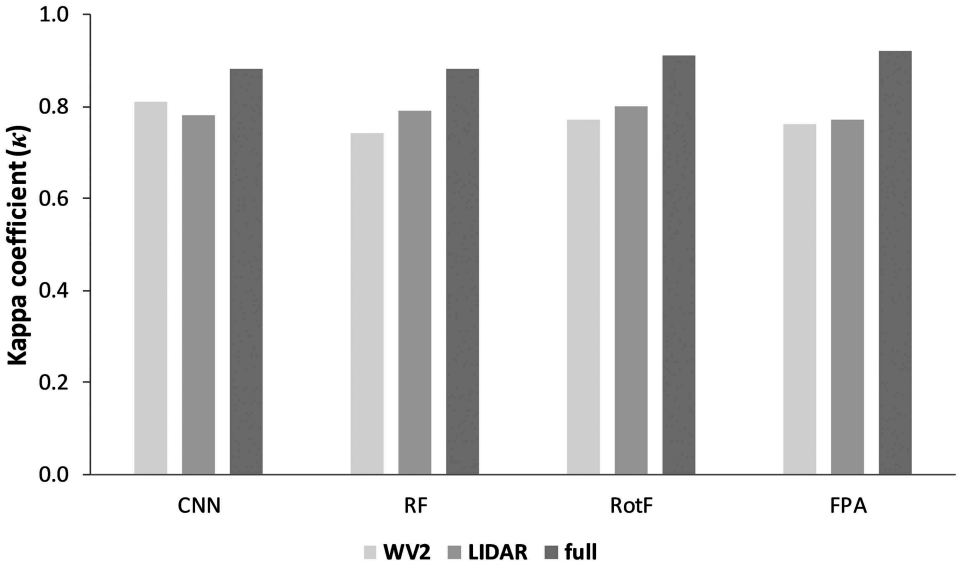


Figure 4. Kappa coefficient (κ) of each classifier and dataset.

FPA and the full dataset, κ_c of 0.81. The reforestation class had a κ_c of 1 with the full dataset for the algorithms FPA and RF, and with the LiDAR dataset for the CNN algorithm. In general, both the LiDAR dataset and the full dataset led to an increase in accuracy of vegetation classes: araucaria, SS_1, SS_2 and reforestation.

3.2. Variable importance

The importance of LiDAR data for vegetation classes discrimination can be better understood with the boxplots of Figure 6. Height metrics such as zmean, zq10, zkurt and zskew (Figure 6b and c), for instance, showed potential to differentiate araucaria class from SS_2, while the CHM (Figure 6(a)) clearly differentiates araucaria and SS_2 from SS_1, which occurs mainly because the latter one has a smaller height.

Intensity LiDAR metrics such as imax and imean (Figure 6(d)) can be useful to separate reforestation from other vegetation classes, since that class presented different range values from native vegetation classes (SS_1, SS_2 and araucaria). However, they are not so useful to discriminate these three classes from each other. In general, it can be seen that intensity metrics showed more potential to discriminate vegetation classes from other land cover classes than vegetation classes among themselves.

Classes with reduced height range, such as bare soil, field and crop, had a better performance with the WV2 dataset than with the LiDAR dataset. This result can be noticed in the classification map of Figure 5 (detail 'C'), and it can be better understood when we compare the boxplots of Figure 6, that show the LiDAR metrics, and the spectral bands of WV2 data (Figure 6(e)). In some cases, the field and crop classes tend to present similar range values of bare soil in the LiDAR boxplots, while in the spectral bands it was noticed that this class can be separated from the others with the last three bands of the WV2 sensor. Figure 6(e) also showed that these three bands seem to be more important to separate land cover classes, mainly when we deal with several vegetation classes. It is

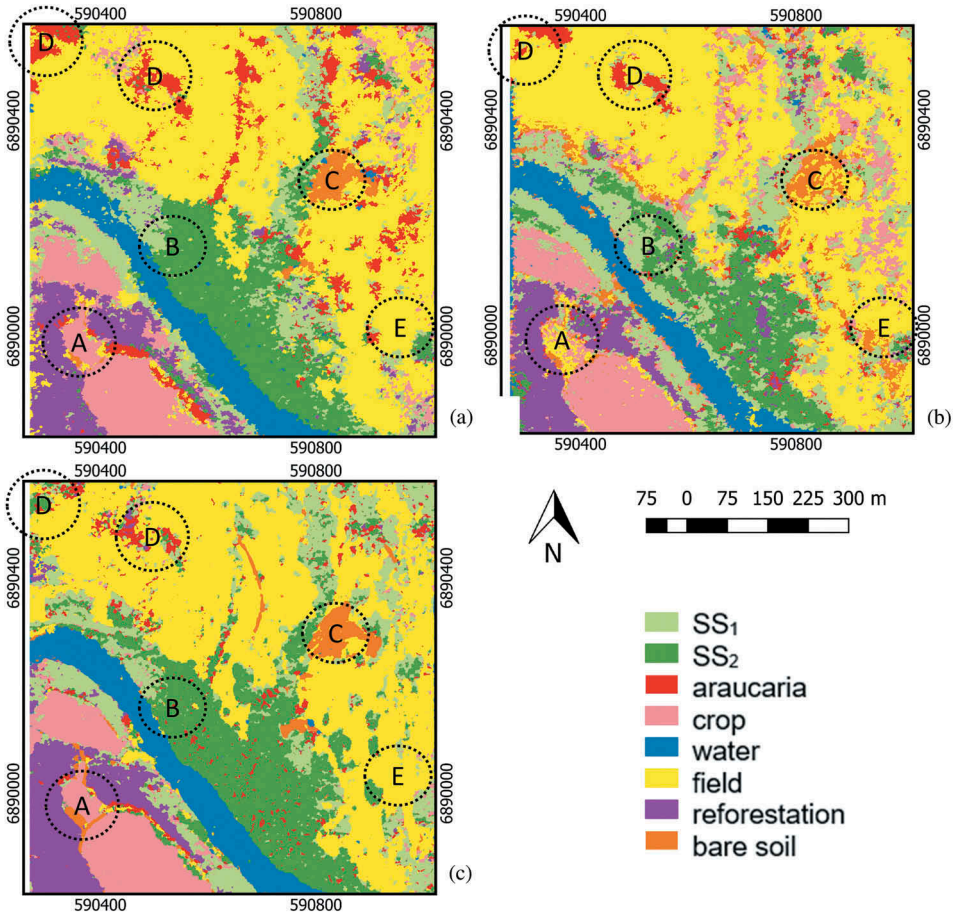
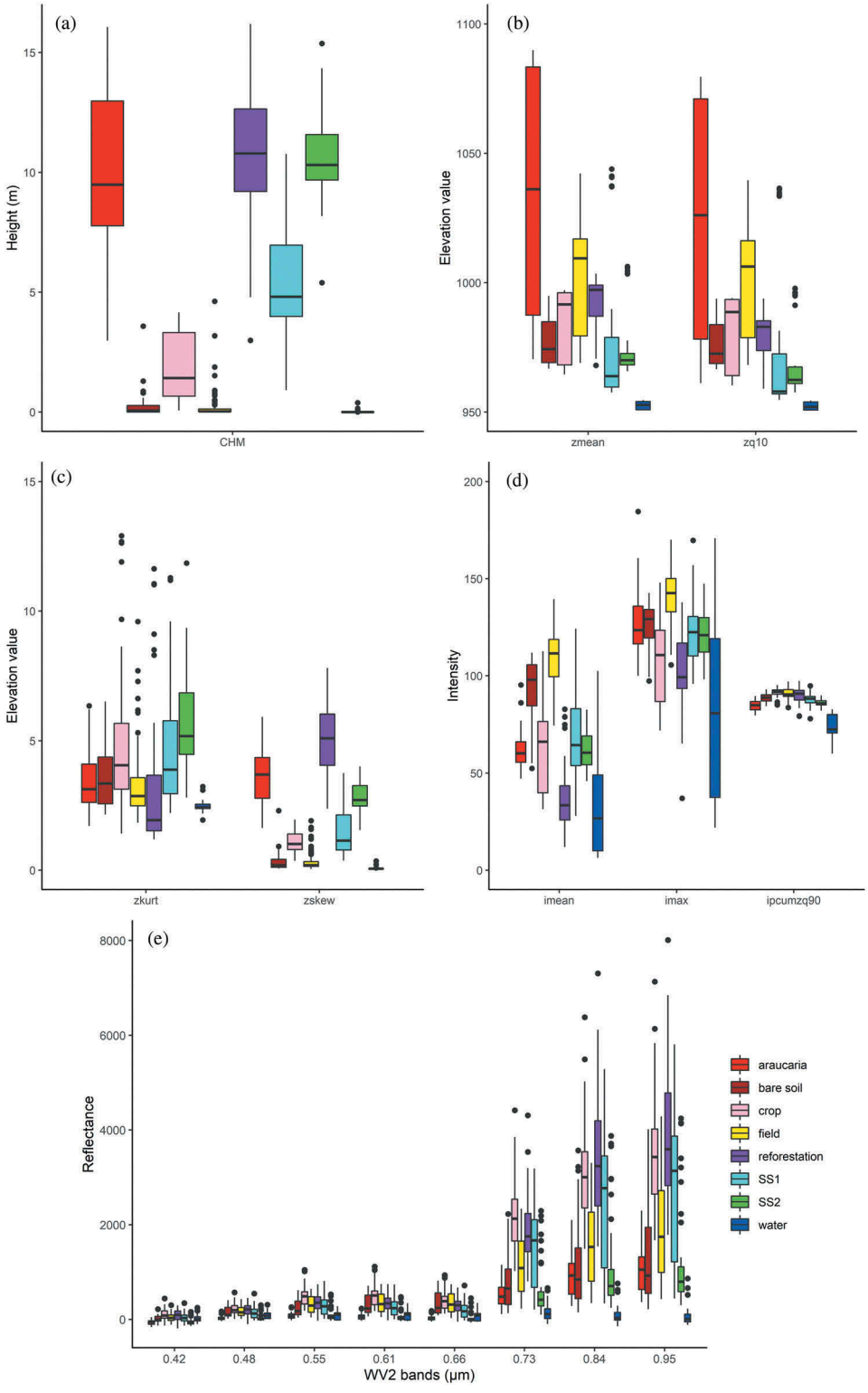


Figure 5. Thematic map of the best classification result obtained by each dataset: a) full dataset with FPA classifier; b) LiDAR dataset with RotF classifier; c) WV2 dataset with CNN classifier. Note in dashed circles: A) crop area wrongly classified as field in (b); B) parts of SS_1 wrongly classified as SS_2 in (a) and (c); C) parts of bare soil wrongly classify as field in (b); D) araucaria fragments wrongly classified as SS_1 or SS_2 in (c); E) parts of field area wrongly classified as SS_1 and crop or bare soil in (b).

Table 4. Conditional kappa (κ_c) index of each class according to the dataset and classifier. The best results for each class are highlighted.

	WV2				LiDAR				full			
Class/Dataset	CNN	RF	RotF	FPA	CNN	RF	RotF	FPA	CNN	RF	RotF	FPA
field	0.85	0.78	0.79	0.69	0.77	0.73	0.65	0.5	**0.91**	0.83	0.76	0.86
SS_2	0.53	0.77	0.74	0.74	0.56	0.65	0.7	0.88	0.69	0.9	**0.98**	0.9
SS_1	0.56	0.5	0.57	0.71	0.71	0.61	0.71	0.68	0.79	0.7	0.76	**0.81**
water	0.98	**1**	**1**	0.84	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**1**
crop	**0.98**	0.74	0.87	0.77	0.9	0.86	0.79	0.73	0.91	0.89	**0.98**	0.91
bare soil	**1**	0.98	**1**	0.87	0.81	0.91	0.91	0.85	**1**	**1**	**1**	**1**
araucaria	0.8	0.57	0.65	0.55	0.58	0.82	0.85	0.93	0.85	0.82	0.96	0.86
reforestation	0.95	0.59	0.65	0.75	**1**	0.78	0.95	0.84	0.9	**1**	0.9	**1**


(a)
Height (m)
CHM
(b)
Elevation value
zmean
zq10
(c)
Elevation value
zkurt
zskew
(d)
Intensity
imean
imax
ipcumzq90
(e)
Reflectance
WV2 bands (μm)
0.42
0.48
0.55
0.61
0.66
0.73
0.84
0.95
araucaria
bare soil
crop
field
reforestation
SS1
SS2
water

possible to observe that in the visible region (0.42–0.66 μm) the vegetation classes present very similar surface reflectance values, while in the red-edge (0.71–0.75) and NIR region (0.77–1.04 μm) these classes are more separable from one another.

Figure 7 shows the importance of the 20 most important features according to the JM distance and the MDG. It can be noticed that different features were considered more important for each method. According to JM, the CHM, five LiDAR metrics, three spectral bands, four VIs and eight textural features are among the most important features. On the other hand, for the MDG all the spectral bands of WV2 are among the most important features, and besides the CHM, only two LiDAR metrics appear in this ranking. Similarly to the JM distance, nine textural features are in the ranking, but the VIs were not among the most important features according to MDG. Both rankings confirm what was seen in Figure 6 regarding the importance of the metrics derived from the LiDAR data, red-edge and NIR bands. The CHM, zskew, and zmean are among the most important metrics for both methods. In the same way, the red-edge, two NIR bands and textural metrics derived from NIR2 band were amid the important features.

4. Discussion

In this study, the best results for all classifiers were obtained with the integration of WV2 and LiDAR data (full dataset), with an increase of 7% to 16% in the κ when compared to the use of optical or LiDAR data alone. Other authors also have shown the potential of combining optical images with LiDAR data for forest applications. Dalponte, Bruzzone, and Gianelle (2008) and Jones, Coops, and Sharma (2010) found that LiDAR combined with hyperspectral data can improve classification accuracy by about 2%, with a greater improvement for particular tree species. Dalponte, Bruzzone, and Gianelle (2012), when classifying tree species and forest types with machine learning algorithms, observed that the contribution of LiDAR was higher when combined with multispectral data than with hyperspectral images, obtaining an increase in accuracy of approximately 14%. Regarding multispectral data, Ke, Quackenbush, and Im (2010) showed that the combination of LiDAR metrics with QuickBird bands noticeably improved tree species classification at the segment level. Reese et al. (2014) classified an alpine vegetation area combining Satellite Probatoire d'Observation de la Terre (SPOT)-5 satellite data, height derivatives and LiDAR data metrics. They observed that the overall classification accuracy of twelve classes improved by approximately 14.2% when LiDAR data were combined with SPOT satellite data.

Comparing the use of the WV2 or the LiDAR dataset separately, the general performance of the classifiers varied very little, showing that in this case, if we had only the

Figure 6. Boxplots of each class with LiDAR features and WV2 spectral bands: a) LiDAR CHM; b) LiDAR height metrics zmean and zq10; c) LiDAR height metrics zkurt and zskew; d) LiDAR intensity metrics imean, imax and ipcumzq90; e) WV2 spectral bands. Note: CHM = Canopy Height Model; zmean = height mean; zq10 = 10th percentile of height distribution; zkurt = kurtosis of height distribution; zskew = skewness of height distribution; imean = intensity mean of all points; imax = maximum of all intensity points; ipcumzq90 = percentage of intensity returned below the 90$^{th.}$

CHM = Canopy Height Model; zmean = height mean; zq10 = 10th percentile of height distribution; zkurt = kurtosis of height distribution; zskew = skewness of height distribution; imean = intensity mean of all points; imax = maximum of all intensity points; ipcumzq90 = percentage of intensity returned below the 90th.

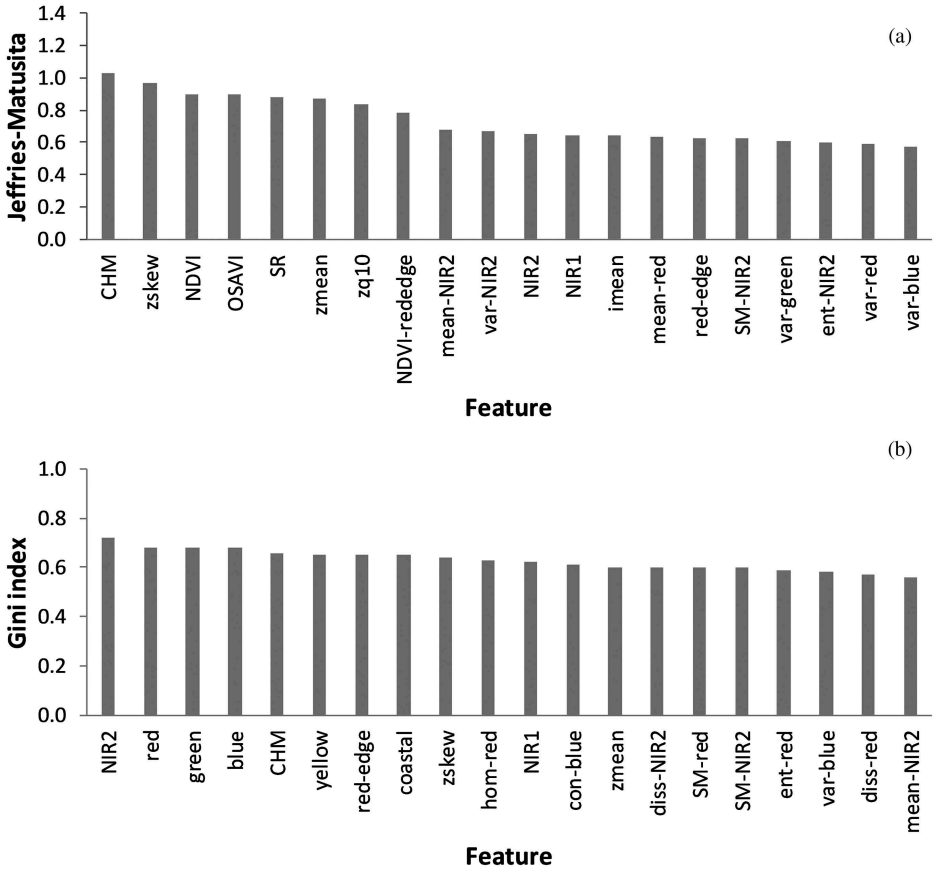


Figure 7. Twenty most important features according to the Jeffries-Matusita distance (a) and the Gini index value (b).

active or the passive sensor, we would reach similar results. The option for employing either the WV2 or the LiDAR data, in the case both are not available, depends on the classes of interest. In general, classifications using only LiDAR data achieved better performance than optical data alone for vegetation-related classes (SS_1, SS_2, araucaria and reforestation), while classes such as field, bare soil and crop obtained better results with optical data. Mainly with respect to the height metrics from LiDAR data, classes spectrally similar, such as the two succession stages, araucaria tree species and reforestation with exotic tree species, could be better separated because of their differences in structure and height. Araucaria class tends to present a greater range in height metrics, because this species is prevalent throughout the study area and presents both pioneer and late succession characteristics (Soares et al. 1979). Besides that, in the study area araucaria trees were located both sparse in the field and in the middle of forest fragments. When located amidst the forest, araucaria tends to be a dominant species in the canopy, and the use of LiDAR data, which provide information on the forest vertical structure, increased the accuracy of this class. The same happened for the discrimination of the two succession stages, since SS_1, besides differences in the species composition, has a shorter height and a smaller number of vertical strata than SS_2. This was also pointed by Castillo

et al. (2012), that used LiDAR data from LVIS sensor to classify three succession stages of vegetation in a tropical rainforest of Costa Rica. They demonstrated that changes in forest vertical structure (such as height) associated with the main succession stages (early, intermediate and late) of the forest can be effectively identified with LiDAR data. Other studies as Cho et al. (2012), Naidoo et al. (2012) and Asner et al. (2008) also reported that the tree height derived from LiDAR was an important variable for mapping tree species in environments such as savannas and tropical forests.

Regarding the other LiDAR metrics, some studies have also emphasized the height percentiles, canopy-return-density and pulse-return-intensity, to describe vertical structure of individual trees and classify tree-species (Andersen, Reutebuch, and McGaughey 2006; Kim et al. 2009; Ørka, Næsset, and Bollandsås 2009; Dalponte, Bruzzone, and Gianelle 2012). Ørka, Næsset, and Bollandsås (2009) used the metrics of height percentiles and canopy-return-densities individually to classify tree-species of Norway spruce (*Picea abies* L.) and Birch (*Betula* sp.), obtaining 74% and 77% of overall accuracies, respectively. Vaughn, Moskal, and Turnblom (2012) used the metrics of point distribution and return intensity to identify five tree-species in the Pacific Northwest United States, resulting in an overall accuracy of 79.2%.

With the exception of the reforestation class, which presents different range values in relation to other vegetation classes, the intensity LiDAR metrics were less useful to discriminate these classes. It was also observed in Figure 7 that, for both feature ranking methods, the height metrics were more important than intensity metrics to discriminate the classes in general. It was mainly because the intensity is a metric that varies as a function of the interaction of the signal with the surface of the target (Reutebuch, Andersen, and Mcgaughey 2005), and vegetation types tend to present similar characteristics regarding this aspect. According to Wing et al. (2012), previous researches have indicated that intensity is useful for differentiating between echoes originating in vegetation and those from other sources, which can be observed in this study, but not to discriminate vegetation classes among themselves.

Regarding the WV2 dataset, it was observed that the bands situated in the red-edge and the NIR regions tend to better discriminate all the classes, mainly the vegetation physiognomies (Immitzer, Atzberger, and Koukal 2012). In a research by Dalponte, Bruzzone, and Gianelle (2012), they obtained lower accuracy in the classification of forest species using multispectral rather than hyperspectral data. The authors argued that one of the reasons was that the multispectral sensor used (in that case, GeoEye) did not have spectral bands in the red-edge region. The lack of the red-edge band of the multispectral data used by Dalponte, Bruzzone, and Gianelle (2012) may have impacted their results. When combining LiDAR features and GeoEye multispectral data, they reached a κ of 0.66 for the classification at the tree level, and 0.69 at the forest types level, which were inferior to the results of our study, that reached a κ of 0.92. Many studies point out the latest generation multispectral sensors, such as WV2 and WV3, with bands located in these spectral ranges (red-edge and NIR), which can bring significant improvements in the classification of tree species (Cho et al. 2012; Alonzo, Bookhagen, and Roberts 2014; Ferreira et al. 2016, 2019). It can also be noticed in Figures 6(e) and 7 the importance of the NIR range (0.77 to 1.04 μm) for vegetation classification. In this region, the SS_1 presented relatively higher surface reflectance values and a greater spectral response than the other vegetation classes. The higher reflectance values are somehow expected,

since with the canopy roughness increase in later succession stages, the reflectance in this spectral region decreases due to the mutual shading either of forest strata or of dominant treetops, which project themselves to the upper part of the forest canopy (Ponzoni, Shimabukuro, and Kuplich 2012). The great variability of the values of this class can be ascribed to the species types present in the area. In some places, shrub vegetation is mostly found, while in others grass or even small trees are prevailing.

Besides the LiDAR metrics and NIR bands, the JM distance (Figure 7) also pointed the VIs as relevant features for land cover classification, which was also previously reported in other studies (Naidoo et al. 2012; Sothe et al. 2017; Maschler, Atzberger, and Immitzer 2018). In the study of Naidoo et al. (2012), the NDVI was scored as one of the most important VIs to classify eight savanna tree species with hyperspectral data. Sothe et al. (2017) reported that the VIs derived from Operational Land Imager (OLI) data had great importance for successional forest stages classification. Maschler, Atzberger, and Immitzer (2018) found great improvement in tree species classification accuracies when VIs and principal components were aggregated to the classifications. In fact, band ratios and VIs generated using different band combinations have advantages for species differentiation and biomass estimation, because these features can reduce Bi-directional Reflectance Distribution Function (BRDF) errors and do not saturate as quickly as single band data (Colgan et al. 2012).

The general performance of all machine learning algorithms was similar in most cases. Among the ensemble methods algorithms, the RotF and FPA generally were better than RF. RotF algorithm had a remarkable performance for two vegetation classes (SS_2 and araucaria), for the crop class, and also the best general result for the LiDAR dataset; FPA, on the other hand, reached the best general accuracy of all algorithms tested, and for one vegetation class (SS_1). Adnan and Islam (2017) compared FPA with other methods, such as RF and RotF. The authors clarified that FPA, conversely to RF, uses the whole feature space, and hence, generates trees with high individual precision and more balanced decision forests compared with other prominent decision forest algorithms. Kavzoglu, Colkesen, and Yomralioglu (2015), when testing algorithms for land cover classification, reported that RotF algorithm had a better performance than other algorithms, such as RF and *k*-nearest neighbour. According to the authors, although RotF is a variant of RF method, it produced more accurate classification results due to its training strategy, in which each individual tree in the ensemble is trained on the complete dataset in a rotated feature space derived from PCA transformations (Kavzoglu, Colkesen, and Yomralioglu 2015).

The CNN achieved the best overall accuracy for the WV2 dataset. Only recently, CNN has started to be explored for the purpose of remote sensing data classification, and most experiments in this respect have generally used hyperspectral data (Makantasis et al. 2015; Yue et al. 2015; Yu, Jia, and Xu 2017; Li Y. et al. 2017; Wang et al. 2017; Gao, Samsung, and Jia 2018). To our knowledge, there are no studies involving the integration of data from multiple sources in order to classify forest environments with CNN. As pointed by Zhang, L. et al. (2016) and Li, W. et al. (2017), in the case of big datasets, the hundreds of bands along the spectral dimension (network inputs) require a large number of parameters to be estimated, which can result in an increased computational cost and the overfitting problem, especially when the training sample set is small. In spite of that, with the WV2 dataset the CNN reached the best result even using only the original WV2 bands, while for other algorithms further features such as VIs and texture metrics were employed

to achieve a comparable performance. This is an advantage of the CNN method, which does not need a prior step of feature extraction before the classification process. The CNN also reached the greatest κ_c for araucaria in the WV2 dataset (Table 3). Li et al. (2017) showed the great potential of this method for tree species detection with multispectral data. They proposed a CNN-based structure to detect and account for palm trees with QuickBird images in Malaysia, comparing it with conventional ANN, local maximum filter (LMF) and template matching (TMPL) methods. They found that CNN performed better than the other methods for the three regions studied, reaching a global accuracy of up to 98.77%.

When the LiDAR dataset was employed, the CNN reached a slightly worse result than other classifiers. It can be due to the use of metrics associated with the intensity and height extracted from LiDAR data, instead of the raw point cloud. According to He et al. (2018), most of the existing methods do not extract the informative features from LiDAR-derived rasterized digital surface models (LiDAR-DSM) data in a deep manner. Qi et al. (2016); (2017) mentioned that 3D data have attracted less attention due to more costly acquisition/processing and their less intuitive and less conventional representation formats, which demand non-trivial pre-processing techniques to discretize the data and make them usable for deep-learning methods. In this sense, some approaches can be explored in future works, such as voxel spaces to create representations that can drive and be processed by a CNN 3D (Maturana and Scherer 2015; Wu et al. 2015) or even the use of morphological and multiattribute profiles to extract information from LiDAR data (He et al. 2018).

Considering the processing time of the classifiers, the ensemble methods algorithms showed to be faster than CNN, which represents an advantage of such methods, especially when dealing with datasets of bulky dimension. Huang, Davis, and Townshend (2002) noted that the higher accuracy of ANNs compared with decision trees algorithms was offset by the larger computational cost. Moreover, ensemble methods algorithms have less hyperparameters to be defined, which eases and encourages their replication in further areas. Although all the machine learning classifiers have parameters to be optimized, the parameter values used in this work provided an output similar or even more accurate when compared to other studies (e.g. Dalponte, Bruzzone, and Gianelle 2012; Kavzoglu, Colkesen, and Yomralioglu 2015; Immitzer, Vuolo, and Atzberger 2016; Raczko and Zagajewski 2017). According to Maxwell, Warner, and Fang (2018), some algorithms have been reported to be robust to parameters settings, and machine learning may still outperform parametric classifiers, such as maximum likelihood, even with little effort to tune hyperparameters.

5. Conclusion

This study explored the use of airborne LiDAR data and multispectral WV2 data separately and their combination (full dataset) for the purpose of land cover classification. As expected, the full dataset led to a significant increase in classifications accuracies for all tested algorithms, confirming the potential of multisource data for this end. When analysed solely, the LiDAR dataset tended to discriminate better the spectrally similar classes, such as araucaria tree species, the two succession stages and reforestation, due to the marked height and structural differences among these classes. In turn, the WV2 dataset presented better performance for classes with less structure variation, such as field, crop and bare soil.

The machine and deep-learning algorithms showed a great potential for classifying datasets coming from different sources, such as multispectral images and LiDAR data. All the classifiers had a similar performance, and none of them attained the best accuracy for all land cover classes. In general terms, FPA had the highest κ for the full dataset (0.92), CNN for the WV2 dataset (0.81), and RotF for the LiDAR dataset (0.8).

The findings of this research are relevant for the conservation of this severely threatened biome, which includes the tree level mapping of endangered species, and can optimize the mapping and monitoring of its forest remnants. As directions for future work, we envisage expanding the study area and applying the tested methods in similar vegetation physiognomies, exploring variations in parameters and further features and datasets settings. We also intend to explore other CNN architectures in order to extract deep features directly from the LiDAR point cloud as well as to test other deep-learning-based classification approaches.

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References

Abadi, M., A. Agarwal, P. Barham, E. Brevdo, Z. Chen, C. Citro, G. S. Corrado, et al. 2015. "TensorFlow: Large-scale Machine Learning on Heterogeneous Systems." Software available from tensorflow.org

Achanccaray, P. M. D., V. A. Ayma, L. Jimenez, S. Garcia, P. N. Happ, R. Q. Feitosa, and A. Plaza. 2014. "A Free Software Tool for Automatic Tuning of Segmentation Parameters." *South-Eastern European Journal of Earth Observation and Geomatics* 3: 707–712. doi:10.3390/ijgi4042292.

Adnan, N., and Z. Islam. 2017. "Forest PA: Constructing a Decision Forest by Penalizing Attributes Used in Previous Trees." *Expert Systems with Applications* 89: 389–403. doi:10.1016/j.eswa.2017.08.002.

Alonzo, M., B. Bookhagen, and D. A. Roberts. 2014. "Urban Tree Species Mapping Using Hyperspectral and LiDAR Data Fusion." *Remote Sensing of Environment* 148: 70–83. doi:10.1016/j.rse.2014.03.018.

Andersen, H. E., S. E. Reutebuch, and R. J. McGaughey. 2006. "A Rigorous Assessment of Tree Height Measurements Obtained Using Airborne LiDAR and Conventional Field Methods." *Canadian Journal of Remote Sensing* 32: 355–366. doi:10.5589/m06-030.

Asner, G. P., D. E. Knapp, T. Kennedy-Bowdoin, M. O. Jones, R. E. Martin, J. Boardman, and R. F. Hughes. 2008. "Invasive Species Detection in Hawaiian Rainforests Using Airborne Imaging Spectroscopy and LiDAR." *Remote Sensing of Environment* 112 (5): 1942–1955. doi:10.1016/j.rse.2007.11.016.

Axelsson, P. 2000. "DEM Generation from LASER Scanner Data Using Adaptive TIN Models." *International Archives of Photogrammetry and Remote Sensing* 33. part. B4: 110-117.

Baatz, M., and A. Schäpe. 2000. "Multiresolution Segmentation — An Optimization Approach for High Quality Multi-scale Image Segmentation." In *Angewandte Geographische Informations-Verarbeitung XII*, edited by J. Strobl, T. Blaschke, and G. Griesebner, 12–23. Karlsruhe: Wichmann Verlag.

Baatz, M., and M. Mimler. 2007. "Objetos De Imagens Iniciais Como Suporte Para a Extração De Objetos De Interesse." In *Sensoriamento Remoto E SIG Avançados* 2. edited by, T. Blaschke and H. Kux.. São Paulo: Oficina de textos. ch. 16: 221-230.

Baldeck, C. A., G. P. Asner, R. E. Martin, C. B. Anderson, D. E. Knapp, J. R. Kellner, and S. J. Wright. 2015. "Operational Tree Species Mapping in a Diverse Tropical Forest with Airborne Imaging Spectroscopy." *PloS One* 10: e0118403. doi:10.1371/journal.pone.0118403.

Belgiu, M., and L. Dragut. 2016. "Random Forest in Remote Sensing: A Review of Applications and Future Directions." *ISPRS Journal of Photogrammetry and Remote Sensing* 114: 24–31. doi:10.1016/j.isprsjprs.2016.01.011.

Breiman, L. 2001. "Random Forests." *Machine Learning* 45 (1): 5–32. doi:10.1023/A:1010933404324.

Breiman, L., J. Friedman, R. Olshen, and C. Stone. 1985. *Classification and Regression Trees*. CA, USA: Wadsworth International Group.

Bruzzone, L., L. Carlin, L. Alparone, S. Baronti, A. Garzelli, and F. Nencini. 2006. "Can Multiresolution Fusion Techniques Improve Classification Accuracy?" Presented at Proceedings of SPIE 6365, Image and Signal Processing for Remote Sensing XII, 636509, Stockholm, Sweden.

Castillo, M., B. Rivard, A. Sánchez-Azofeifa, J. Calvo-Alvarado, and R. Dubayah. 2012. "LiDAR Remote Sensing for Secondary Tropical Dry Forest Identification." *Remote Sensing of Environment* 121: 132–143. doi:10.1016/j.rse.2012.01.012.

Cho, M. A., R. Mathieu, G. P. Asner, L. Naidoo, J. Van Aardt, A. Ramoelo, P. Debba, et al. 2012. "Mapping Tree Species Composition in South African Savannas Using an Integrated Airborne Spectral and LiDAR System." *Remote Sensing of Environment* 125: 214–226. doi:10.1016/j.rse.2012.07.010.

Clark, M. L., and D. A. Roberts. 2012. "Species-level Differences in Hyperspectral Metrics among Tropical Rainforest Trees as Determined by a Tree-based Classifier." *Remote Sensing* 4 (6): 1820–1855. doi:10.3390/rs4061820.

Colgan, M. S., C. ;. A. Baldeck, J. B. Féret, and G. P. Asner. 2012. "Mapping Savanna Tree Species at Ecosystem Scales Using Support Vector Machine Classification and BRDF Correction on Airborne Hyperspectral and LiDAR Data." *Remote Sensing* 4: 3462–3480. doi:10.3390/rs4113462.

Congalton, R., and K. Green. 2009. *Assessing the Accuracy of Remotely Sensed Data: Principles and Practices*. Boca Raton: CRC/Taylor & Francis. 183p.

Dalponte, M., H. O. Ørka, T. Gobakken, D. Gianelle, and E. Næsset. 2013. "Tree Species Classification in Boreal Forests with Hyperspectral Data." *IEEE Transactions on Geoscience and Remote Sensing* 51: 2632–2645. doi:10.1109/TGRS.2012.2216272.

Dalponte, M., L. Bruzzone, and D. Gianelle. 2008. "Fusion of Hyperspectral and LiDAR Remote Sensing Data for Classification of Complex Forest Areas." *IEEE Transactions on Geoscience and Remote Sensing* 46: 1416–1427. doi:10.1109/TGRS.2008.916480.

Dalponte, M., L. Bruzzone, and D. Gianelle. 2012. "Tree Species Classification in the Southern Alps Based on the Fusion of Very High Geometrical Resolution multispectral/hyperspectral Images and LiDAR Data." *Remote Sensing of Environment* 123: 258–270. doi:10.1016/j.rse.2012.03.013.

Debes, C. A., R. Merentitis, J. Heremans, N. Hahn, T. Frangiadakis, L. van Kasteren, R. Wenzhi, et al. 2014. "Hyperspectral and LiDAR Data Fusion: Outcome of the 2013 GRSS Data Fusion Contest." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 7 :2405–2418. doi:10.1109/JSTARS.2014.2305441.

Digital Globe. 2013. *DigitalGlobe Core Imagery Product Guide V1.1*. Available at: <https://geomatics.planet.com/upload/digitalglobe/DigitalGlobe%20Core%20Imagery%20Products%20Guide.pdf> Accessed 10 Mar 2018.

Driss, S. B., M. Soua, and M. Akil. 2017. "A Comparison Study between MLP and Convolutional Neural Network Models for Character Recognition." SPIE Conference on Real-Time Image and Video Processing, Anaheim, CA, United States. SPIE Proceedings 10223, Real-Time Image and Video Processing 2017. doi: 10.1117/12.2262589.

Duro, D. C., S. E. Franklin, and M. G. Dubé. 2012. "Multi-scale Object-based Image Analysis and Feature Selection of Multi-sensor Earth Observation Imagery Using Random Forests." *International Journal of Remote Sensing* 33: 4502–4526. doi:10.1080/01431161.2011.649864.

Feitosa, R. Q., R. S. Ferreira, C. M. Almeida, F. F. Camargo, and G. A. O. P. Costa. 2010. "Similarity Metrics for Genetic Adaptation of Segmentation Parameters." *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* XXXVIII: 150–155.

Féret, J., and G. P. Asner. 2013. "Tree Species Discrimination in Tropical Forests Using Airborne Imaging Spectroscopy." *IEEE Transactions on Geoscience and Remote Sensing* 51: 73–84. doi:10.1109/TGRS.2012.2199323.

Ferreira, M. P., F. H. Wagner, L. E. O. C. Aragão, Y. E. Schimabukuro, and C. R. de Souza Filho. 2019. "Tree Species Classification in Tropical Forests Using Visible to Shortwave Infrared WorldView-3 Images and Texture Analysis." *ISPRS Journal of Photogrammetry and Remote Sensing* 149: 119–131. doi:10.1016/j.isprsjprs.2019.01.019.

Ferreira, M. P., M. Zortea, D. C. Zanotta, Y. E. Shimabukuro, and C. R. de Souza Filho. 2016. "Mapping Tree Species in Tropical Seasonal Semi-deciduous Forests with Hyperspectral and Multispectral Data." *Remote Sensing of Environment* 179: 66–78. doi:10.1016/j.rse.2016.03.021.

Fieber, K. D., I. J. Davenport, J. M. Ferryman, R. J. Gurney, J. P. Walker, and J. M. Hacker. 2013. "Analysis of Full-waveform LiDAR Data for Classification of an Orange Orchard Scene." *ISPRS Journal of Photogrammetry and Remote Sensing* 82: 63–82. doi:10.1016/j.isprsjprs.2013.05.002.

Fukushima, K. 1980. "Neocognitron: A Self-organizing Neural Network Model for A Mechanism of Pattern Recognition Unaffected by Shift in Position." *Biological Cybernetics* 36: 193–202. doi:10.1007/BF00344251.

Gao, Q., L. Samsung, and X. Jia. 2018. "Hyperspectral Image Classification Using Convolutional Neural Networks and Multiple Feature Learning." *Remote Sensing* 10: 299. doi:10.3390/rs10020299.

Ghosh, A., E. F. Fassnacht, P. K. Joshi, and B. Koch. 2014. "A Framework for Mapping Tree Species Combining Hyperspectral and LiDAR Data: Role of Selected Classifiers and Sensor across Three Spatial Scales." *International Journal of Applied Earth Observation and Geoinformation* 26: 49–63. doi:10.1016/j.jag.2013.05.017.

Goodfellow, I., Y. Bengio, and A. Courville. 2016. *Deep Learning*. MIT press, Cambridge.

Hall, M., E. Frank, G. Holmes, B. Pfahringer, P. Reutemann, and I. H. Witten. 2009. "The Weka Data Mining Software: an Update." *Sigkdd Explorations Newsletter* 11 (1): 10-18.

Haralick, R. M., K. Shanmugam, and I. H. Dinstein. 1973. "Textural Features for Image Classification." *IEEE Transactions on Systems Man and Cybernetics* 3: 610–621. doi:10.1109/TSMC.1973.4309314.

Hastie, T. J., R. J. Tibshirani, and J. H. Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. New York: Springer.

He, X., A. Wang, P. Ghamisi, G. Li, and Y. Chen. 2018. "LiDAR Data Classification Using Spatial Transformation and CNN." *IEEE Geoscience and Remote Sensing Letters*. doi:10.1109/LGRS.2018.2868378.

Heinzel, J., and B. Koch. 2011. "Exploring Full-waveform LiDAR Parameters for Tree Species Classification." *International Journal of Applied Earth Observation and Geoinformation* 13: 152–160. doi:10.1016/j.jag.2010.09.010.

Higuchi, P., A. C. da Silva, T. de S. Ferreira, S. T. de Souza, J. P. Gomes, K. M. Silva, K. F. Dos Santos, C. Linke, and P. da S. Paulino. 2012. "Influência De Variáveis Ambientais Sobre O Padrão Estrutural E Florístico Do Componente Arbóreo, Em Um Fragmento De Floresta Ombrófila Mista Montana Em Lages, SC." *Ciência Florestal* 22 (1): 79–90. doi:10.5902/198050985081.

Huang, C., L. S. Davis, and J. R. G. Townshend. 2002. "An Assessment of Support Vector Machines for Land Cover Classification." *International Journal of Remote Sensing* 23 (4): 725–749. doi:10.1080/01431160110040323.

Immitzer, M., C. Atzberger, and T. Koukal. 2012. "Tree Species Classification with Random Forest Using Very High Spatial Resolution 8-band WorldView-2 Satellite Data." *Remote Sensing* 4 (9): 2661–2693. doi:10.3390/rs4092661.

Immitzer, M., F. Vuolo, and C. Atzberger. 2016. "First Experience with Sentinel-2 Data for Crop and Tree Species Classifications in Central Europe." *Remote Sensing* 8: 166. doi:10.3390/rs8030166.

Isenburg, M. 2012. "LAStools: Software for Rapid LiDAR Processing." (http://www.cs.unc.edu/~isenburg/lastools/) doi: 10.1094/PDIS-11-11-0999-PDN.

Joly, C. A., J. P. Metzger, and M. Tabarelli. 2014. "Experiences from the Brazilian Atlantic Forest: Ecological Findings and Conservation Initiatives." *New Phytologist* 204 (3): 459–473. doi:10.1111/nph.12989.

Jones, T. G., N. C. Coops, and T. Sharma. 2010. "Assessing the Utility of Airborne Hyperspectral and LiDAR Data for Species Distribution Mapping in the Coastal Pacific Northwest Canada." *Remote Sensing of Environment* 114: 2841–2852. doi:10.1016/j.rse.2010.07.002.

Kavzoglu, T., I. Colkesen, and T. Yomralioglu. 2015. "Object-based Classification with Rotation Forest Ensemble Learning Algorithm Using Very-high-resolution WorldView-2 Image." *Remote Sensing Letters* 6 (11): 834–843. doi:10.1080/2150704X.2015.1084550.

Ke, Y., L. J. Quackenbush, and J. Im. 2010. "Synergistic Use of Quickbird Multispectral Imagery and LiDAR Data for Object-based Forest Species Classification." *Remote Sensing of Environment* 114: 1141–1154. doi:10.1016/j.rse.2010.01.002.

Kim, S., R. J. McGaughey, H. E. Andersen, and G. Schreuder. 2009. "Tree Species Differentiation Using Intensity Data Derived from Leaf-on and Leaf-off Airborne Laser Scanner Data." *Remote Sensing of Environment* 113: 1575–1586. doi:10.1016/j.rse.2009.03.017.

Kingma, D. P., and J. L. Ba. 2015. "Adam: A Method for Stochastic Optimization." International Conference on Learning Representations, 1–13, San Diego, USA.

Kuncheva, L. I., and J. J. Rodríguez. 2007. "An Experimental Study on Rotation Forest Ensembles." In *Multiple Classifier Systems. MCS 2007. Lecture Notes in Computer Science*, edited by, M. Haindl, J. Kittler, and F. Roli. Vol. 4472 459–468. Berlin/ Heidelberg,Germany: Springer. ISBN 978-3-54-072481-0.

Leckie, D. G., F. A. Gougeon, S. Tinis, T. Nelson, C. N. Burnett, and D. Paradine. 2005. "Automated Tree Recognition in Old Growth Conifer Stands with High Resolution Digital Imagery." *Remote Sensing of Environment* 94 (3): 311–326. doi:10.1016/j.rse.2004.10.011.

Li, W., F. Haohuan, L. Yu, and A. Cracknell. 2017. "Deep Learning Based Oil Palm Tree Detection and Counting for High-Resolution Remote Sensing Images." *Remote Sensing* 9 (1): 22. doi:10.3390/rs9010022}.

Li, Y., H. Zhang, and Q. Shen. 2017. "Spectral–Spatial Classification of Hyperspectral Imagery with 3D Convolutional Neural Network." *Remote Sensing* 9 (67). doi:10.3390/rs9010067.

Louppe, G., L. Wehenkel, A. Sutera, and P. Geurts. 2013. "Understanding Variable Importances in Forests of Randomized Trees." *NIPS'13 Proceedings of the 26th International Conference on Neural Information Processing Systems* 1: 431–439, Lake Tahoe, USA.

Lu, D., G. Li, E. Moran, and W. Kuang. 2014. "A Comparative Analysis of Approaches for Successional Vegetation Classification in the Brazilian Amazon." *GIScience and Remote Sensing* 51: 695–709. doi:10.1080/15481603.2014.983338.

Lu, D., and Q. Weng. 2007. "A Survey of Image Classification Methods and Techniques for Improving Classification Performance." *International Journal of Remote Sensing* 28: 823–870. doi:10.1080/01431160600746456.

Ma, L., T. Fu, T. Blaschke, M. Li, D. Tiede, Z. Zhou, and D. Chen. 2017. "Evaluation of Feature Selection Methods for Object-Based Land Cover Mapping of Unmanned Aerial Vehicle Imagery Using Random Forest and Support Vector Machine Classifiers." *ISPRS International Journal of Geo-Information* 6 (2): 51. doi:10.3390/ijgi6020051.

Makantasis, K., K. Karantzalos, A. Doulamis, and N. Doulamis. 2015. "Deep Supervised Learning for Hyperspectral Data Classification through Convolutional Neural Networks." *Proceedings of IEEE International Geoscience and Remote Sensing Symposium*, Milan, Italy, 4959–4962.

Mas, J. F., and J. J. Flores. 2008. "The Application of Artificial Neural Networks to the Analysis of Remotely Sensed Data." *International Journal of Remote Sensing* 29 (3): 617–663. doi:10.1080/01431160701352154.

Maschler, J., C. Atzberger, and M. Immitzer. 2018. "Individual Tree Crown Segmentation and Classification of 13 Tree Species Using Airborne Hyperspectral Data." *Remote Sensing* 10: 1218. doi:10.3390/rs10081218.

Maturana, D., and S. Scherer. 2015. "Voxnet: A 3d Convolutional Neural Network for Real-time Object Recognition." *In 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems* (IROS), 922–928, Hamburg, Germany.

Maxwell, A. E., T. A. Warner, and F. Fang. 2018. "Implementation of Machine-learning Classification in Remote Sensing: An Applied Review." *International Journal of Remote Sensing* 39: 2784–2817. doi:10.1080/01431161.2018.1433343.

Mountrakis, G., J. Im, and C. Ogole. 2011. "Support Vector Machines in Remote Sensing: A Review." *ISPRS Journal of Photogrammetry and Remote Sensing* 66: 247–259. doi:10.1016/j.isprsjprs.2010.11.001.

Naidoo, L., M. A. Cho, R. Mathieu, and G. Asner. 2012. "Classification of Savanna Tree Species, in the Greater Kruger National Park Region, by Integrating Hyperspectral and LiDAR Data in a Random Forest Data Mining Environment." *ISPRS Journal of Photogrammetry and Remote Sensing* 69: 167–179. doi:10.1016/j.isprsjprs.2012.03.005.

Nelder, J., and R. Mead. 1965. "A Simplex Method for Function Minimization." *The Computer Journal* 7 (4): 308–313. doi:10.1093/comjnl/7.4.308.

Omer, G., O. Mutanga, E. M. Abdel-Rahman, and E. Adam. 2015. "Performance of Support Vector Machines and Artificial Neural Network for Mapping Endangered Tree Species Using Worldview-2 Data in Dukuduku Forest, South Africa." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 8 (10): 4825–4840. doi:10.1109/JSTARS.2015.2461136.

Ørka, H. O., E. Næsset, and O. M. Bollandsås. 2009. "Classifying Species of Individual Trees by Intensity and Structure Features Derived from Airborne Laser Scanner Data." *Remote Sensing of Environment* 113: 1163–1174. doi:10.1016/j.rse.2009.02.002.

Pasupa, K., and W. A. Sunhem. 2016. "Comparison between Shallow and Deep Architecture Classifiers on Small Dataset." Proceedings of 8th International Conference on Information Technology and Electrical Engineering (ICITEE), Yogyakarta, Indonesia, 16707994.

Pont-Tuset, J., and F. Marques. 2013. "Measures and Meta-Measures for the Supervised Evaluation of Image Segmentation." Paper presented at *IEEE Conference on Computer Vision and Pattern Recognition- CVPR*: 2131–2138. doi:10.1109/CVPR.2013.277.

Ponzoni, F. J., Y. E. Shimabukuro, and T. M. Kuplich. 2012. *Sensoriamento Remoto Da Vegetação*. 2 ed. São Paulo: Oficina de Textos.

Pouteau, R., J. Y. Meyer, R. Taputuarai, and B. Stoll. 2012. "Support Vector Machines to Map Rare and Endangered Native Plants in Pacific Islands Forests." *Ecological Informatics* 9: 37–46. doi:10.1016/j.ecoinf.2012.03.003.

Pu, R. L., and S. Landry. 2012. "A Comparative Analysis of High Spatial Resolution IKONOS and WorldView-2 Imagery for Mapping Urban Tree Species." *Remote Sensing of Environment* 124: 516–533. doi:10.1016/j.rse.2012.06.011.

Qi, C. R., H. Su, K. Mo, and L. J. Guibas. 2017. "Pointnet: Deep Learning on Point Sets for 3d Classification and Segmentation." *Computer Vision and Pattern Recognition* (CVPR), IEEE, 1, 4: 652–660.

Qi, C. R., H. Su, M. Nießner, A. Dai, M. Yan, and L. J. Guibas. 2016. "Volumetric and Multi-view CNNs for Object Classification on 3d Data." *Computer Vision and Pattern Recognition* (CVPR), IEEE 1, 5648–5656. doi: 10.1109/cvpr.2016.609.

Qian, Y., W. Zhou, J. Yan, W. Li, and L. Han. 2014. "Comparing Machine Learning Classifiers for Object-Based Land Cover Classification Using Very High Resolution Imagery." *Remote Sensing* 7: 153–168. doi:10.3390/rs70100153.

Quinlan, R. 1993. *C4.5: Programs for Machine Learning*. San Francisco: Morgan Kaufmann.

Quirita, V. A. A., P. M. D. Achanccaray, R. Q. Feitosa, P. N. Happ, G. A. O. P. Costa, T. Klinger, and C. Heipke. 2016. "Metaheuristics for Supervised Parameter Tuning of Multiresolution Segmentation." *IEEE Geoscience and Remote Sensing Letters* 13 (9): 1364–1368. doi:10.1109/LGRS.2016.2586499.

R Development Core Team. 2013. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. http://www.R-project.org/

Raczko, E., and B. Zagajewski. 2017. "Comparison of Support Vector Machine, Random Forest and Neural Network Classifiers for Tree Species Classification on Airborne Hyperspectral APEX Images." *European Journal of Remote Sensing* 50 (1): 144–154. doi:10.1080/22797254.2017.1299557.

Reese, H., M. Nyström, K. Nordkvist, and H. Olsson. 2014. "Combining Airborne Laser Scanning Data and Optical Satellite Data for Classification of Alpine Vegetation." *International Journal of Applied Earth Observation and Geoinformation* 27: 81–90. doi:10.1016/j.jag.2013.05.003.

Reutebuch, S. E., H. E. Andersen, and R. J. Mcgaughey. 2005. "Light Detection and Ranging (Lidar): An Emerging Tool for Multiple Resource Inventory." *Journal of Forestry, Society of American Foresters* 103 (6): 286–292.

Ribeiro, M. C., J. P. Metzger, A. C. Martensen, F. J. Ponzoni, and M. M. Hirota. 2009. "The Brazilian Atlantic Forest: How Much Is Left, and How Is the Remaining Forest Distributed? Implications for Conservation." *Biological Conservation* 142: 1141–1153. doi:10.1016/j.biocon.2009.02.021.

Richards, J. A. 1993. *Remote Sensing Digital Image Analysis: An Introduction*. Berlin: Springer-Verlag.

Rodríguez, J. J., L. I. Kuncheva, and C. J. Alonso. 2006. "Rotation Forest: A New Classifier Ensemble Method." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 2810. doi:10.1109/TPAMI.2006.211.

Rodriguez-Galiano, V. F., B. Ghimire, J. Rogan, M. Chica-Olmo, and J. P. Rigol-Sanchez. 2012. "An Assessment of the Effectiveness of a Random Forest Classifier for Landcover Classification." *ISPRS Journal of Photogrammetry and Remote Sensing* 67: 93–104. doi:10.1016/j.isprsjprs.2011.11.002.

Rodriguez-Galiano, V. F., and M. Chica-Rivas. 2014. "Evaluation of Different Machine Learning Methods for Land Cover Mapping of a Mediterranean Area Using Multi-seasonal Landsat Images and Digital Terrain Models." *International Journal of Digital Earth* 7 (6): 492–509. doi:10.1080/17538947.2012.748848.

Roussel, J. R., D. Auty, F. de Boissieu, and A. S. Meador. 2018. "lidR: Airborne LiDAR Data Manipulation and Visualization for Forestry Applications, Version 1.4.1." accessed 23 March 2018 https://cran.r-project.org/web/packages/lidR/index.html

Singh, M., D. Evans, B. S. Tan, and C. S. Nin. 2015. "Mapping and Characterizing Selected Canopy Tree Species at the Angkor World Heritage Site in Cambodia Using Aerial Data." *PloS One* 10 (4): e0121558. doi:10.1371/journal.pone.0121558.

Soares, R. V. 1979. "Considerações Sobre a Regeneração Natural Da Araucaria Angustifólia." *Floresta* 10: 11–18. doi:10.5380/rf.v10i2.6249.

Song, X., Z. Duan, and X. Jiang. 2012. "Comparison of Artificial Neural Networks and Support Vector Machine Classifiers for Land Cover Classification in Northern China Using a SPOT-5 HRG Image." *International Journal of Remote Sensing* 33: 3301–3320. doi:10.1080/01431161.2011.568531.

Sothe, C., C. M. de Almeida, V. Liesenberg, and M. B. Schimalski. 2017. "Evaluating Sentinel-2 and Landsat-8 Data to Map Sucessional Forest Stages in a Subtropical Forest in Southern Brazil." *Remote Sensing* 9: 838. doi:10.3390/rs9080838.

Stojanova, D., P. Panov, V. Gjorgjioski, A. Kobler, and S. Džeroski. 2010. "Estimating Vegetation Height and Canopy Cover from Remotely Sensed Data with Machine Learning." *Ecological Informatics* 5: 256–266. doi:10.1016/j.ecoinf.2010.03.004.

Storn, P., and K. Price. 1997. "Differential Evolution – A Simple and Efficient Heuristic for Global Optimization over Continuous Spaces." *Journal of Global Optimization* 11: 341–359. doi:10.1023/A:1008202821328.

Tsendbazar, N. E. 2011. "Object Based Image Analysis of Geo-eye VHR Data to Model above Ground Carbon Stock in Himalayan Mid-hill Forests, Nepal." M.Sc. Thesis, Geo-Information Science and Earth Observation, University of Twente, Enschede, The Netherlands.

Vaughn, N. R., L. M. Moskal, and E. C. Turnblom. 2012. "Tree Species Detection Accuracies Using Discrete Point LiDAR and Airborne Waveform LiDAR." *Remote Sensing* 4: 377–403. doi:10.3390/rs4020377.

Vibrans, A. C., L. Sevegnani, A. L. Gasper, and D. V. Lingner. 2013. *Inventário Florístico Florestal De Santa Catarina: Floresta Ombrófila Mista*. Blumenau: Edifurb, v.3.

Wang, J., C. Luo, H. Huang, H. Zhao, and S. Wang. 2017. "Transferring Pre-Trained Deep CNNs for Remote Scene Classification with General Features Learned from Linear PCA Network." *Remote Sensing* 9 (225). doi:10.3390/rs9030225.

Whiteside, T. 2011. Multiscale vegetation characterization of tropical savanna using object-based image analysis. Ph.D. Dissertation, Charles Darwin University, Melbourne.

Wing, B. M., M. W. Ritchie, K. Boston, W. B. Cohen, A. Gitelman, and M. J. Olsen. 2012. "Prediction of Understory Vegetation Cover with Airborne LiDAR in an Interior Ponderosa Pine Forest." *Remote Sensing of Environment* 124: 730–741. doi:10.1016/j.rse.2012.06.024.

Wu, Z., S. Song, A. Khosla, F. Yu, L. Zhang, X. Tang, and J. Xiao. 2015. "3d Shapenets: A Deep Representation for Volumetric shapes." In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 1912–1920, Boston, USA.

Xia, J., P. Du, X. He, and J. Chanussot. 2013. "Hyperspectral Remote Sensing Image Classification Based on Rotation Forest." *IEEE Geoscience and Remote Sensing Letters* 13884355: 239–243. doi:10.1109/LGRS.2013.2254108.

Yu, Q., P. Gong, N. Clinton, G. Biging, M. Kelly, and D. Schirokauer. 2006. "Object-based Detailed Vegetation Classification with Airborne High Spatial Resolution Remote Sensing Imagery." *Photogrammetric Engineering and Remote Sensing* 72: 799–811. doi:10.14358/PERS.72.7.799.

Yu, S., S. Jia, and C. Xu. 2017. "Convolutional Neural Networks for Hyperspectral Image Classification." *Neurocomputing* 219: 88–98. doi:10.1016/j.neucom.2016.09.010.

Yue, J., W. Zhao, S. Mao, and H. Liu. 2015. "Spectral–Spatial Classification of Hyperspectral Images Using Deep Convolutional Neural Networks." *Remote Sensing Letters* 6: 468–477. doi:10.1080/2150704X.2015.1047045.

Zhang, L., L. Zhang, and B. Du. 2016. "Deep Learning for Remote Sensing Data: A Technical Tutorial on the State of the Art," *IEEE Geoscience and Remote Sensing Magazine* 4: 22–40. doi: 10.1109/MGRS.2016.2540798.

Zhang, Z., R. Zang, G. Wang, and X. Huang. 2016. "Classification of Landscape Types Based on Land Cover, Successional Stages and Plant Functional Groups in a Species-rich Forest in Hainan Island, China." *Tropical Conservation Science* 9: 135–152. doi:10.1177/194008291600900107.